

NOP (00)		Cycles: 2	Size: 1
Implied			
Cycle	R/W		Desc.
-X.2	Read PC		Store as OpCode.
1.1			Inc. PC .
1.2			
2.1			
2.2	Read PC		Store as OpCode.
+X.1			

TCALL (01)	Cycles: 8	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		
2.2		
3.1		
3.2	Write to u8(SP +0) \$0100	Push PC .H onto stack.
4.1		
4.2	Write to u8(SP -1) \$0100	Push PC .L onto stack.
5.1		Dec. SP by 2.
5.2		
6.1		
6.2	Read \$FFDE	Store as PC .L.
7.1		
7.2	Read \$FFDF	Store as PC .H.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		

SET1 (02)	Cycles: 4	Size: 2
Direct Page Bit d.b (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Operand = (Operand & ~\$01) \$01.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

BBS0 (03)	Cycles: 5-7	Size: 3
Direct Page Bit Relative d.b, r		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (Operand & (1 << 0)) == (1 << 0).
3.2		
4.1		
4.2	Read PC	Store as Operand.
5.1		Inc. PC . If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = PC + i8(Operand). PC .L = Address.L.
6.2		
7.1		PC .H = Address.H.
7.2	Read PC	Store as OpCode.
+X.1		

OR (04)	Cycles: 3	Size: 2
Direct Page d (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
3.2	Read PC	Store as OpCode.
+X.1		

OR (05)	Cycles: 4	Size: 3
Absolute !a (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		

OR (06)	Cycles: 3	Size: 1
Indirect X (X) (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1	Read (X ((PSW & P flag) << 3))	Inc. PC .
1.2		
2.1		
2.2		Store as Operand.
3.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
3.2		Store as OpCode.
+X.1		

OR (07)	Cycles: 6	Size: 2
X-Indexed Indirect [d+X] (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		Address = ((X + Operand) & \$FF) ((PSW & P flag) << 3).
3.2	Read Address	Store as Pointer.L.
4.1		
4.2	Read ((Address + 1) & \$FF) ((PSW & P flag) << 3)	Store as Pointer.H.
5.1		
5.2	Read Pointer	Store as Operand.
6.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Read PC	Store as OpCode.
+X.1		

OR (08)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . A = Operand. N flag = (A & \$80), Z flag = ! A .
2.2	Read PC	Store as OpCode.
+X.1		

OR (09)	Cycles: 6	Size: 3
Direct Page to Direct Page dd, ds (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand0.
3.1		
3.2	Read PC	Store as Operand.
4.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
4.2	Read Address	Store as Operand1.
5.1		Operand0 = Operand1. N flag = (Operand0 & \$80), Z flag = !Operand0.
5.2	Write to Address	Write Operand0.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

OR1 (0A)	Cycles: 5	Size: 3
Absolute Boolean Bit m.b		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read (Address & \$1FFF)	Store as Operand.
4.1		Bit = Address >> 13. C flag = ((Operand & (1 << Bit)) != 0).
4.2		
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

ASL (0B)	Cycles: 4	Size: 2
Direct Page d (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		C flag = (Operand & \$80). Operand <= 1. N flag = (Operand & \$80), Z flag = !Operand.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

ASL (0C)	Cycles: 5	Size: 3
Absolute !a (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		C flag = (Operand & \$80). Operand <= 1. N flag = (Operand & \$80), Z flag = !Operand.
4.2	Write to Address	Write Operand.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

PUSH (0D)		Cycles: 4	Size: 1
Implied			
Cycle	R/W		Desc.
-X.2	Read PC		Store as OpCode.
1.1			Inc. PC .
1.2	Read PC		Discard.
2.1			
2.2	Write to u8(SP +0) \$0100		Push PSW onto stack.
3.1			Dec. SP by 1.
3.2			
4.1			
4.2	Read PC		Store as OpCode.
+X.1			

TSET1 (0E)	Cycles: 6	Size: 3
Absolute !a (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		Tmp = (A - Operand). Operand = A . Z flag = (Tmp == 0), N flag = (Tmp & \$80).
4.2	Read Address	Discard.
5.1		
5.2	Write to Address	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

BRK (0F)	Cycles: 8	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		
2.2	Write to u8(SP +0) \$0100	Push PC .H onto stack.
3.1		
3.2	Write to u8(SP -1) \$0100	Push PC .L onto stack.
4.1		
4.2	Write to u8(SP -2) \$0100	Push PSW onto stack.
5.1		Dec. SP by 3. I flag = 0, X flag = 1.
5.2		
6.1		
6.2	Read \$FFDE	Store as PC .L.
7.1		
7.2	Read \$FFDF	Store as PC .H.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		

BPL (10)	Cycles: 2-4	Size: 2
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC . Set Jump to (PSW & \$80) == \$00.
1.2		Store as Operand.
2.1		Inc. PC . If !Jump, end instruction (next half-cycle is 4.2).
2.2		
3.1		Address = PC + i8(Operand).
		PC .L = Address.L.
3.2		
4.1		PC .H = Address.H.
4.2	Read PC	Store as OpCode.
+X.1		

TCALL1 (11) Cycles: 8		Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		
2.2		
3.1		
3.2	Write to u8(SP +0) \$0100	Push PC .H onto stack.
4.1		
4.2	Write to u8(SP -1) \$0100	Push PC .L onto stack.
5.1		Dec. SP by 2.
5.2		
6.1		
6.2	Read \$FFDC	Store as PC .L.
7.1		
7.2	Read \$FFDD	Store as PC .H.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		

CLR1 (12)	Cycles: 4	Size: 2
Direct Page Bit d.b (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Operand = (Operand & ~\$01) \$00.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

BBC0 (13)	Cycles: 5-7	Size: 3
Direct Page Bit Relative d.b, r		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (Operand & (1 << 0)) == (0 << 0).
3.2		
4.1		
4.2	Read PC	Store as Operand.
5.1		Inc. PC . If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = PC + i8(Operand). PC .L = Address.L.
6.2		
7.1		PC .H = Address.H.
7.2	Read PC	Store as OpCode.
+X.1		

OR (14)	Cycles: 4	Size: 2
X-Indexed Direct Page d+X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		Address = ((X + Operand) & \$FF) ((PSW & P flag) << 3).
3.2	Read Address	Store as Operand.
4.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		

OR (15)	Cycles: 5	Size: 3
X-Indexed Absolute !a+X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2		
4.1		Pointer = (X + Address).
4.2	Read Pointer	Store as Operand.
5.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Read PC	Store as OpCode.
+X.1		

OR (16)	Cycles: 5	Size: 3
Y-Indexed Absolute !a+Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2		
4.1		Pointer = (Y + Address).
4.2	Read Pointer	Store as Operand.
5.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Read PC	Store as OpCode.
+X.1		

OR (17)	Cycles: 6	Size: 2
Indirect Y-Indexed [d]+Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		Address = (Operand ((PSW & P flag) << 3)).
3.2	Read Address	Store as Pointer.L.
4.1		
4.2	Read ((Address + 1) & \$FF) ((PSW & P flag) << 3)	Store as Pointer.H.
5.1		Address = (Y + Pointer).
5.2	Read Address	Store as Operand.
6.1		A = Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Read PC	Store as OpCode.
+X.1		

OR (18)	Cycles: 5	Size: 3
Immediate Data to Direct Page d, #i (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand0.
2.1		Inc. PC .
2.2	Read PC	Store as Operand.
3.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
3.2	Read Address	Store as Operand1.
4.1		Operand0 = Operand1. N flag = (Operand0 & \$80), Z flag = !Operand0.
4.2	Write to Address	Write Operand0.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

OR (19)	Cycles: 5	Size: 1
Indirect Page to Indirect Page		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		
2.2	Read (Y ((PSW & P flag) << 3))	Store as Operand0.
3.1		
3.2	Read (X ((PSW & P flag) << 3))	Store as Operand1.
4.1		Operand0 = Operand1. N flag = (Operand0 & \$80), Z flag = !Operand0.
4.2	Write to (X ((PSW & P flag) << 3))	Write Operand0.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

DECW (1A)	Cycles: 6	Size: 2
Direct Page d (Read/Modify/Write 16)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Underflow = (Operand == 0). Operand -= 1. LowZero = (Operand == 0).
3.2	Write to Address	Write Operand.
4.1		
4.2	Read ((Address + 1) & \$FF) ((PSW & P flag) << 3)	Store as Operand.
5.1		Operand -= Underflow. N flag = (Operand & \$80), Z flag = (LowZero && !Operand).
5.2	Write to ((Address + 1) & \$FF) ((PSW & P flag) << 3)	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

ASL (1B)	Cycles: 5	Size: 2
X-Indexed Direct Page d+X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		Address = ((X + Operand) & \$FF) ((PSW & P flag) << 3).
3.2	Read Address	Store as Operand.
4.1		C flag = (Operand & \$80). Operand <= 1. N flag = (Operand & \$80), Z flag = !Operand.
4.2	Write to Address	Write Operand.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

ASL (1C)	Cycles: 2	Size: 1
Accumulator A		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		C flag = (A & \$80). A <<= 1. N flag = (A & \$80), Z flag = ! A .
2.2	Read PC	Store as OpCode.
+X.1		

DEC (1D)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2		
2.1		X -= 1. N flag = (X & \$80), Z flag = ! X .
2.2		Store as OpCode.
+X.1		

CMP (1E)	Cycles: 4	Size: 3
Absolute !a (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		Tmp = X - Operand. C flag = (Tmp >= 0), N flag = (Tmp & \$80), Z = !Tmp.
4.2	Read PC	Store as OpCode.
+X.1		

JMP (1F)	Cycles: 6	Size: 3
Absolute X-Indexed Indirect [!a+X]		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		
3.2		
4.1		Pointer = (X + Address).
4.2	Read Pointer	Store as PC .L.
5.1		
5.2	Read Pointer + 1	Store as PC .H.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

CLRP (20)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2		
2.1		PSW = (PSW & ~\$20).
2.2		Store as OpCode.
+X.1		

TCALL1 (21) Cycles: 8		Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		
2.2		
3.1		
3.2	Write to u8(SP +0) \$0100	Push PC .H onto stack.
4.1		
4.2	Write to u8(SP -1) \$0100	Push PC .L onto stack.
5.1		Dec. SP by 2.
5.2		
6.1		
6.2	Read \$FFDA	Store as PC .L.
7.1		
7.2	Read \$FFDB	Store as PC .H.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		

SET1 (22)	Cycles: 4	Size: 2
Direct Page Bit d.b (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Operand = (Operand & ~\$02) \$02.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

BBS1 (23)	Cycles: 5-7	Size: 3
Direct Page Bit Relative d.b, r		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (Operand & (1 << 1)) == (1 << 1).
3.2		
4.1		
4.2	Read PC	Store as Operand.
5.1		Inc. PC . If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = PC + i8(Operand). PC .L = Address.L.
6.2		
7.1		PC .H = Address.H.
7.2	Read PC	Store as OpCode.
+X.1		

AND (24)	Cycles: 3	Size: 2
Direct Page d (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .
3.2	Read PC	Store as OpCode.
+X.1		

AND (25)	Cycles: 4	Size: 3
Absolute !a (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		

AND (26)	Cycles: 3	Size: 1
Indirect X (X) (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1	Read (X ((PSW & P flag) << 3))	Inc. PC .
1.2		
2.1		
2.2		Store as Operand.
3.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .
3.2		Store as OpCode.
+X.1		

AND (27)	Cycles: 6	Size: 2
X-Indexed Indirect [d+X] (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		Address = ((X + Operand) & \$FF) ((PSW & P flag) << 3).
3.2	Read Address	Store as Pointer.L.
4.1		
4.2	Read ((Address + 1) & \$FF) ((PSW & P flag) << 3)	Store as Pointer.H.
5.1		
5.2	Read Pointer	Store as Operand.
6.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .
6.2	Read PC	Store as OpCode.
+X.1		

AND (28)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . A &= Operand. N flag = (A & \$80), Z flag = ! A .
2.2	Read PC	Store as OpCode.
+X.1		

AND (29)	Cycles: 6	Size: 3
Direct Page to Direct Page dd, ds (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand0.
3.1		
3.2	Read PC	Store as Operand.
4.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
4.2	Read Address	Store as Operand1.
5.1		Operand0 &= Operand1. N flag = (Operand0 & \$80), Z flag = !Operand0.
5.2	Write to Address	Write Operand0.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

OR1 (2A)	Cycles: 5	Size: 3
Absolute Boolean Bit /m.b		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read (Address & \$1FFF)	Store as Operand.
4.1		Bit = Address >> 13. C flag = ((Operand & (1 << Bit)) == 0).
4.2		
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

ROL (2B)	Cycles: 4	Size: 2
Direct Page d (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Tmp = C flag. C flag = (Operand & \$80). Operand = (Operand << 1) Tmp. N flag = (Operand & \$80), Z flag = !Operand.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

ROL (2C)	Cycles: 5	Size: 3
Absolute !a (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address	Store as Operand.
4.1		Tmp = C flag. C flag = (Operand & \$80). Operand = (Operand << 1) Tmp. N flag = (Operand & \$80), Z flag = !Operand.
4.2	Write to Address	Write Operand.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

PUSH (2D)		Cycles: 4	Size: 1
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1	Read PC Write to u8(SP +0) \$0100 Read PC	Inc. PC .	
1.2		Discard.	
2.1			
2.2		Push A onto stack.	
3.1		Dec. SP by 1.	
3.2			
4.1			
4.2		Store as OpCode.	
+X.1			

CBNE (2E)		Cycles: 5-7	Size: 3
Direct Page Relative d, r			
Cycle	R/W	Desc.	
-X.2	Read PC	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC	Store as Operand.	
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).	
2.2	Read Address	Store as Operand.	
3.1		Set Jump to (A != Operand).	
3.2			
4.1			
4.2	Read PC	Store as Operand.	
5.1		Inc. PC . If !Jump, end instruction (next half-cycle is 7.2).	
5.2			
6.1		Address = PC + i8(Operand).	
		PC .L = Address.L.	
6.2			
7.1		PC .H = Address.H.	
7.2	Read PC	Store as OpCode.	
+X.1			

BRA (2F)	Cycles: 4	Size: 2
Relative r		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC . Set Jump to true.
1.2		Store as Operand.
2.1		Inc. PC . If !Jump, end instruction (next half-cycle is 4.2).
2.2		
3.1		Address = PC + i8(Operand).
		PC .L = Address.L.
3.2		
4.1		PC .H = Address.H.
4.2	Read PC	Store as OpCode.
+X.1		

BMI (30)	Cycles: 2-4	Size: 2
Relative r		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC . Set Jump to (PSW & \$80) == \$80.
1.2		Store as Operand.
2.1		Inc. PC . If !Jump, end instruction (next half-cycle is 4.2).
2.2		
3.1		Address = PC + i8(Operand).
		PC .L = Address.L.
3.2		
4.1		PC .H = Address.H.
4.2	Read PC	Store as OpCode.
+X.1		

TCALL3 (31) Cycles: 8		Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		
2.2		
3.1		
3.2	Write to u8(SP +0) \$0100	Push PC .H onto stack.
4.1		
4.2	Write to u8(SP -1) \$0100	Push PC .L onto stack.
5.1		Dec. SP by 2.
5.2		
6.1		
6.2	Read \$FFD8	Store as PC .L.
7.1		
7.2	Read \$FFD9	Store as PC .H.
8.1		
8.2	Read PC	Store as OpCode.
+X.1		

CLR1 (32)	Cycles: 4	Size: 2
Direct Page Bit d.b (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Operand = (Operand & ~\$02) \$00.
3.2	Write to Address	Write Operand.
4.1		
4.2	Read PC	Store as OpCode.
+X.1		

BBC1 (33)	Cycles: 5-7	Size: 3
Direct Page Bit Relative d.b, r		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Set Jump to (Operand & (1 << 1)) == (0 << 1).
3.2		
4.1		
4.2	Read PC	Store as Operand.
5.1		Inc. PC . If !Jump, end instruction (next half-cycle is 7.2).
5.2		
6.1		Address = PC + i8(Operand). PC .L = Address.L.
6.2		
7.1		PC .H = Address.H.
7.2	Read PC	Store as OpCode.
+X.1		

AND (34)	Cycles: 4	Size: 2
X-Indexed Direct Page d+X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		Address = ((X + Operand) & \$FF) ((PSW & P flag) << 3).
3.2	Read Address	Store as Operand.
4.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .
4.2	Read PC	Store as OpCode.
+X.1		

AND (35)	Cycles: 5	Size: 3
X-Indexed Absolute !a+X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2		
4.1		Pointer = (X + Address).
4.2	Read Pointer	Store as Operand.
5.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Read PC	Store as OpCode.
+X.1		

AND (36)	Cycles: 5	Size: 3
Y-Indexed Absolute !a+Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2		
4.1		Pointer = (Y + Address).
4.2	Read Pointer	Store as Operand.
5.1		A &= Operand. N flag = (A & \$80), Z flag = ! A .
5.2	Read PC	Store as OpCode.
+X.1		

AND (37)	Cycles: 6	Size: 2
Indirect Y-Indexed [d]+Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		Address = (Operand ((PSW & P flag) << 3)).
3.2	Read Address	Store as Pointer.L.
4.1		
4.2	Read ((Address + 1) & \$FF) ((PSW & P flag) << 3)	Store as Pointer.H.
5.1		Address = (Y + Pointer).
5.2	Read Address	Store as Operand.
6.1		A &= Operand. N flag = (A & \$80), Z flag = !A.
6.2	Read PC	Store as OpCode.
+X.1		

AND (38)	Cycles: 5	Size: 3
Immediate Data to Direct Page d, #i (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand0.
2.1		Inc. PC .
2.2	Read PC	Store as Operand.
3.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
3.2	Read Address	Store as Operand1.
4.1		Operand0 &= Operand1. N flag = (Operand0 & \$80), Z flag = !Operand0.
4.2	Write to Address	Write Operand0.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

AND (39)	Cycles: 5	Size: 1
Indirect Page to Indirect Page (X), (Y) (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		
2.2	Read (Y ((PSW & P flag) << 3))	Store as Operand0.
3.1		
3.2	Read (X ((PSW & P flag) << 3))	Store as Operand1.
4.1		Operand0 &= Operand1. N flag = (Operand0 & \$80), Z flag = !Operand0.
4.2	Write to (X ((PSW & P flag) << 3))	Write Operand0.
5.1		
5.2	Read PC	Store as OpCode.
+X.1		

INCW (3A)	Cycles: 6	Size: 2
Direct Page d (Read/Modify/Write 16)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Overflow = (Operand == \$FF). Operand += 1. LowZero = (Operand == 0).
3.2	Write to Address	Write Operand.
4.1		
4.2	Read ((Address + 1) & \$FF) ((PSW & P flag) << 3)	Store as Operand.
5.1		Operand += Overflow. N flag = (Operand & \$80), Z flag = (LowZero && !Operand).
5.2	Write to ((Address + 1) & \$FF) ((PSW & P flag) << 3)	Write Operand.
6.1		
6.2	Read PC	Store as OpCode.
+X.1		

ROL (3B)	Cycles: 5	Size: 2
X-Indexed Direct Page d+X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		Address = ((X + Operand) & \$FF) ((PSW & P flag) << 3).
3.2		Store as Operand.
4.1		Tmp = C flag. C flag = (Operand & \$80). Operand = (Operand << 1) Tmp. N flag = (Operand & \$80), Z flag = !Operand.
4.2		Write Operand.
5.1		
5.2		Store as OpCode.
+X.1		

ROL (3C)	Cycles: 2	Size: 1
Accumulator A		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Discard.
2.1		Tmp = C flag. C flag = (A & \$80). A = (A << 1) Tmp. N flag = (A & \$80), Z flag = ! A .
2.2	Read PC	Store as OpCode.
+X.1		

INC (3D)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2		
2.1		X += 1. N flag = (X & \$80), Z flag = ! X .
2.2		Store as OpCode.
+X.1		

CMP (3E)	Cycles: 3	Size: 2
Direct Page d (Read)		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Operand.
2.1		Inc. PC . Address = (Operand ((PSW & P flag) << 3)).
2.2	Read Address	Store as Operand.
3.1		Tmp = X - Operand. C flag = (Tmp >= 0), N flag = (Tmp & \$80), Z = !Tmp.
3.2	Read PC	Store as OpCode.
+X.1		

CALL (3F)	Cycles: 8	Size: 3
Absolute Call !a		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC	Store as Address.L.
2.1		Inc. PC .
2.2	Read PC	Store as Address.H.
3.1		Inc. PC .
3.2		
4.1		
4.2	Write to u8(SP +0) \$0100	Push PC .H onto stack.
5.1		
5.2	Write to u8(SP -1) \$0100	Push PC .L onto stack.
6.1		Dec. SP by 2.
6.2		
7.1		PC .L = Address.L.
7.2		
8.1		PC .H = Address.H.
8.2	Read PC	Store as OpCode.
+X.1		

SETP (40)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC	Store as OpCode.
1.1		Inc. PC .
1.2		
2.1		PSW = (PSW \$20).
2.2		Store as OpCode.
+X.1		